Fabrication of 2-Nitrophenol sensor based on green synthesized hydroxyapatite

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***Abstract*— We have developed a new electrochemical sensing platform to detect environment pollutant 2-Nitrophenol based on hydroxyapatite nanoparticles (HA Nps) modified glassy carbon electrode. Hydroxyapatite nanoparticles were prepared via microwave irradiation method by utilizing different plant extracts (Eucalyptus, Ocimum Bascilium, Plectranthus Amboincus, and Justicia Adathoda). The structural and thermal behavior of plant extracts assisted HA nanoparticles have been characterized by XRD, FTIR, and TGA. The Electrochemical behavior of 2-Nitrophenol was investigated by cyclic voltammetry (CV), differential pulse voltammetry (DPV) techniques by using pristine HA and plant extracts assisted HA Nps modified GCE. Among them, Plectranthus Amboincus assisted HA Nps modified GCE exhibits excellent electro catalytic activity towards 2-Nitrophenol. Moreover, it has been good sensitive for determination of 2-Nitrophenol over a wide linear range from 0.2µM to 60µM with the lowest limit of detection 90nM. The fabricated sensor has high selectivity towards major interfering compounds such as 3-Nitrophenol, 4-Nitrophenol, 4-Aminophenol, Catechol, and Hydroquinone. In addition, the analytical application of the proposed sensor shows a good agreement with real samples.**

I. INTRODUCTION

2-Nitrophenol is one of the main compounds in phenol families, broadly applied in the manufacturing of explosives, dyes, rubber, pharmaceutical, pesticides, and various industrial applications. Even so, inevitably it can release into the environment in the form of pollutants. They affect plants and animals, it can readily accumulate in the human organisms to cause carcinogenesis, mutagenesis by a nature of its toxicity [1-2]. Numerous diseases like methemoglobinemia, headaches, and nausea are caused by inhaled of the 2-Nitrophenol compound in even at very low concentration. 2-Np is cited in a list of pollutant has given by U.S Environmental Protection Agency (EPA) due to its toxicity and persistence [3]. To develop an effective method for determination of 2-Nitrophenol in the environment is important. Several methods are available for detection of 2-Nitrophenol like UV-Visible fluorescence spectrophotometer [4, 5] high-performance liquid chromatography [6], capillary zone electrophoresis [7] and electrochemical methods [8,9].In recent years, electrochemical methods are good considerable attention due to its easily altering the electrode surface by the physically or chemically. Chemically modified electrode was huge advantages such as easy fabrication, low price, renewable surfaces, time- consuming and long-term stability in different solvents [10]. Hydroxyapatite is a predominant mineral for bone and teeth and it has extreme biocompatibility, ion exchangeability, and having a lot of attracting properties [11, 12]. HA Nps have been synthesized by various methods such as solvated, hydrothermal, sol-gel, chemical precipitation and microwave methods [13]. Among them, microwave method is one of the best way to a synthesis of HA Nps because of the simple procedure with the shortest time. To attain various morphology and minimized agglomeration of the HA Nps by a doping with metal ions or composite with organic, inorganic surfactants and using different biosynthesis routes such as bacteria, fungi, and plant extract. It may help to reduce the crystal size to improve the electrocatalytic property and control morphology [14-16]. Here we report to synthesis HA Nps by using different plant extract like Eucalyptus, Ocimum basicilium , Plectranthus Amboinicus, and Jusiticia Adathoda for the first time to detect the toxic environmental pollutant 2-Nitrophenol.The fabricated sensor shows high electrochemical activity towards the 2-Nitophenol in 0.1M PBS (pH 6.0).The sensor was successfully applied for in the real sample analysis.

II. EXPERIMENTAL

2.1 *Reagents*

2-Nitrophenol, Calcium Nitrate tetra hydrate & Di-Ammonium hydrogen phosphate were purchased from Merk Private Limited, Mumbai. All other chemicals were of analytical reagent

grade and used without any further purification. 0.1M of Phosphate buffer solution was prepared from Na_2 HPO_4 and NaH_2PO_4 (Merck, Mumbai) and the pH was adjusted to 6.0 by adding 0.1M HCl. The deionised water was used throughout the experiments.

2.2 Appratus

Electrochemical measurements were carried out on CHI 609D workstation .A three electrode cell systems was used with a modified glassy carbon electrode as the working electrode. An Ag/AgCl as the reference electrode and a platinum wire electrode as the counter electrode an all experiments were carried out at room temperature. Cyclic voltammograms (CVs) were acquired from at a scan rate 50 mV/s in 0.1M phosphate buffer containing 1mM of 2-Nitrophenol.Electrochemical impedance spectroscopy (EIS) measurements were made by applying ac potential of amplitude 5mV over the dc potential of 250mV in the frequency range Nyquist plots. The value of the charge transfer resistance was determined using Zsimpwin software simulations. The structure and morphology of the hydroxyapatite were characterized by powder X-ray powder diffraction (XRD) using a (XPERT-PRO diffractometer with Cu Kα radiation (1.5406Å) at 40 kV) and field emission scanning electron microscopy (FE-SEM) using ZEISS. Fourier transform infrared (FTIR) were recorded from 4000 $cm^{-1}$to 400 cm^{-1}at room temperature with an (ThermoNichlolate) using KBr pellet technique. Thermogravimetric (TG) analysis of the synthesized powder was performed in a NETZCH STA 4491 instrument between 0–1400^0 C in air at heating rate of 10 ^{0}C min^{-1}.

2.3 *Synthesis of Plant extract assisted - Hydroxyapatite Nanoparticle*

Plectranthus amboinicus extract solution was prepared from a 70g portion of thoroughly washed Plectranthus amboinicus leaves which were finely cut and boiled in deionised water. The resulting extract was filtered and used as a solvent (instead of deionised water) for preparing HA. 0.4M ($CaNO_3.4H_2O$) solution(A) and0.24M$(NH_4)_2HPO_4$ solution (B). Solution B was added slowly into solution A with constant stirring. During the reaction, the pH of the solution was maintained at 10 by using ammonia solution. The mixture was then exposed to microwave radiation (600W) for 20 min, which resulted in a light brown color paste like product. This sample was washed with deionised water and dried at 80° C in a hot air oven. The resultant powder is whitish brown in color. Hereafter the sample was coded as Kv-HA. The above procedure was applied to other three plants like Oscimum basicilium (TS-HA), Justcia adathoda (AD-HA), Eucalyptus (EA-HA) by independently. For comparison we prepare pristine HA by using same microwave method by using deionised water instead of plant extract.

2.4 *Electrode Preparation*

Prior to modification, the glassy carbon electrode (GCE, 3 mm in diameter) was polished with 0.05 mm alumina powder on polishing cloth, rinsed thoroughly with deionised water between each polishing step, sonicated and dried at room temperature. The modified electrodes were prepared by a simple drop casting method. HA nanoparticles were dispersed into deionized water (5 mg/ mL^{-1}) by ultrasonicating for about 30 min to get a stable and uniform suspension. Typically,10 µL of HA nanoparticle suspension was drop casted onto the GCE and allowed to evaporate water at room temperature in air.

III. RESULTS AND DISCUSSION

3.1 Characterization of Synthesized Powder

The XRD patterns of the HA, ES-HA, TS-HA, KV-HA, and AD-HA samples prepared by microwave irradiation method are shown in Fig 1(A).The XRD patterns of all samples are perfectly matched with the standard HA crystallites (ICDD PDF Card No-09-432 of Hexagonal phase) . No characteristics peaks of impurities, such as calcium hydroxide and calcium phosphates were observed, which means that the pure HA phase will be appearing in all plant samples. Crystallite size was calculated by the Debye- Scherer formula ($D=K\lambda/\beta_{1/2}\cos\theta$) λ is a wavelength, $\beta_{1/2}$ is a full width at half maximum (FWHM), θ is a diffraction angle, K is shape factor constant. The average crystal size of HA and ES-HA, TS-HA, KV-HA, AD-HA are 30.1, 23.4, 23.5, 14.62, and 25.68nm.When compare with all HA Nps samples, then the KV-HA Nps crystal size is more decrease due to the participation of plectranthus amboincus plant leaves extract and it may act as a capping agent in the synthesis process of HA Nps to against aggregation. Fig 1(B) shows the FT-IR spectra of pure HA and ES-HA, TS-HA, KV-HA, AD-HA samples. A sharp band at 471, 961cm^{-1} were assigned to stretching vibration of PO_4^{3-} respectively. The doublet peaks 602 and 564 cm^{-1} corresponds to bending modes of phosphate ions. A large band at 3568, 3572 cm^{-1} can be assigned to OH stretching vibration modes respectively. In HA peak 1634 cm^{-1} is assigned to bending mode of adsorbed water. But, In KV-HA peak 1614 cm^{-1} is mainly attributed to amide group it indicates the confirmation of plant extracts on synthesized sample .The carbonated peak appeared at 1384 cm^{-1} is related to CO_2 group adsorbed from atmosphere during the synthesis process. The Intensity of peaks 564 and 1384 cm^{-1}were

decreased significantly due to the presence of plant extracts. Fig.2 shows Thermal analyses of both the pristine HA, TS-HA, KV-HA, and AD-HA were carried out in order to understand the role of the all plant extract assisted HA NPs. Thermo gravimetric (TGA) curves were recorded on powdered HA Nps between 0–1400 ^{o} C in air at a heating rate of 10° C min^{-1} . As for results, the TGA curves show four main weight losses in the 200-250 ^{o} C, 400-600 ^{o} C, 620-830 ^{o} C and 1000-1200° C. The weight loss below 250 ^{o} C is due to the evaporation of physically and chemically entrapped water molecules and its significantly higher for TS-HA and KV-HA when compared with pristine HA. Next three decomposition can be obtained for organic constituents, removal of ammonia and nitrate, oxygen from the HA NPs. Subsequently, there is a weight loss of pristine HA and all plant extract assisted HA Nps were 13% and 11% respectively. Notably pristine HA sample shows higher weight loss when compared to the other HA samples. This result indicated the highest thermal stability of the TS-HA, KV-HA samples which could be attributed to its smaller particle size. Fig. 3 (A & B) shows FE-SEM micrographs spectra of the pristine HA and KV-HA samples. The HA crystallizes in the form of layered platelets shaped contain capsule like nanoparticles. Whereas, Plectranthus amboinicus(KV-HA) assisted hydroxyapatite forms as a flower blossoms structure. The long chain of organic compounds is present in Plectranthus amboinicus extract seem to affect the homogeneous distribution of HA crystallites and breaking the layered plates to the formation of flower shape morphology.

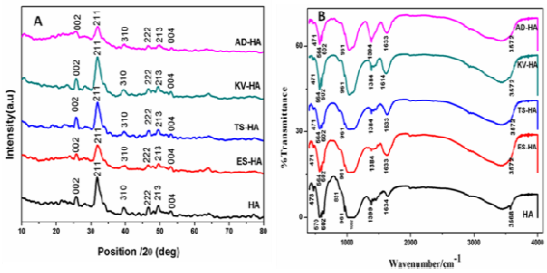

Fig .1. (A) XRD pattern of HA, ES-HA,TS-HA, KV-HA, AD-HA) K V-HA **B)** FT-IR spectra of HA, ES-HA,TS-HA, KV-HA, AD-HA.

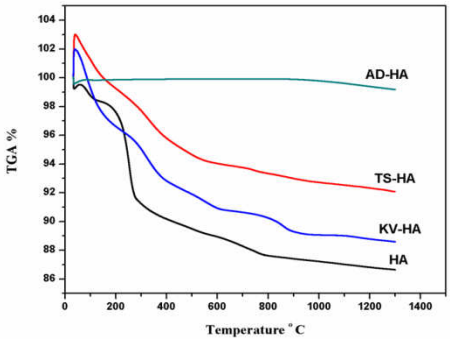

Fig 2 Shows TGA curves of HA, TS-HA, KV-HA, AD-HA

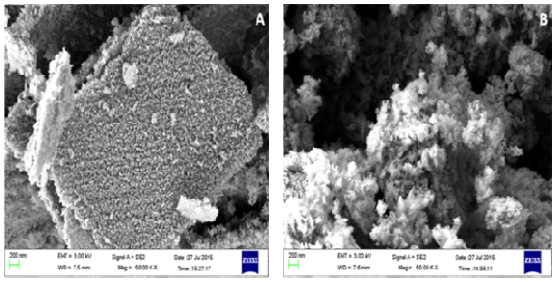

Fig 3. SEM images of **(A)** pristine HA and (**B**) KV-HA

3.2 Electrochemical investigation of 2-Nitrophenol

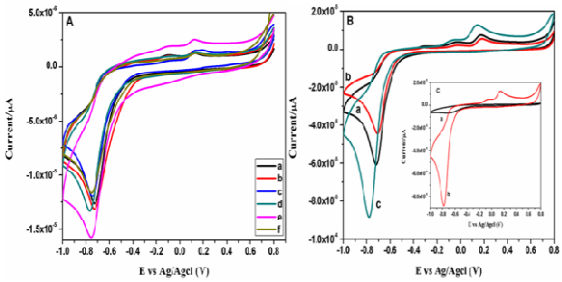

Fig 4. (A) Cyclicvoltammogram of 0.1mM 2-Np in 0.1M PBS (pH 6.0) at a) Bare GCE, b) pure HA ,c) ES-HA, d) TS-HA, e) KV-HA, f) AD-HA modified GCE , at scan rate 50mV s^{-1} .(**B**) Cyclicvoltammogram of 1mM of 2-Np in 0.1M PBS (pH 6.0) at a) Bare GCE, b)HA/GCE, c) KV-HA/GCE (**C**) Cyclicvoltammogram of absence(a) and presence (b) of 1mM of 2-Np in PBS (pH6.0) at Kv-HA/GCE (Inset).

Cyclic voltammetry was used to investigate the electrochemical behavior of 2-Np at different modified electrodes in pH 6.0 phosphate buffer solution at a scan rate 50 mV/s. Fig. 4A shows the CVs of 1mM 2-Np in PBS at bare GCE (curve a), HA/GCE (curve b), ES-HA/GCE (curve c), TS-HA/GCE (curve d), KV-HA/GCE (curve e), AD-HA/GCE (curve f) respectively. Noticeably, a small defined oxidation wave and large reduction peak appeared for all modified electrodes. The large cathodic peak shows the reduction of 2NP to 2-(hydroxyamino) phenol. The small anodic peak indicates the oxidation of 2-(hydroxyamino) phenol to 2- Nitrosophenol. When compared with all modified GCE, the KV-HA (curve e) modified GCE exhibited significantly enhanced redox current. This can mainly be attributed to the presence of plectranthus amboincus plant leaves extract. In order to investigate the catalytic activities of KV-HA/GCE towards the electrochemical response in high concentration (1mM) of 2- Np in 0.1 M PBS (pH= 6.0) at a scan rate of 50 mV/s. When compared with bare GCE and HA/GCE, the fabricated sensor shows a better redox peak. The KV-HA /GCE did not show any redox peak in the absence (black line) of 2-Np indicating that the KV-

HA/GCE non-electroactive in plank PBS When 1mM of 2-Np was added into 0.1M PBS, a pair redox peaks were observed on KV-HA/GCE. It is no doubt that the redox peaks should be attributed to the electrochemical behavior of 2-Np.

3.3 *Effect of Scan rate*

Fig. 5A shows the effect of scanning rate on cyclic voltammograms of 2-Np at the KV-HA/GCE in PBS at pH 6.0. That the oxidation and reduction peaks increased linearly with the increase of scan rates from 100 to 500mVs^{-1} and exhibited a linear relation to the square root of the scan rate, which implies that the redox reaction of 2-Np at the KV-HA /GCE is a surface adsorption-controlled process. The above results also demonstrated that the proposed sensor provide a facilitating electron transfer towards 2-Np.

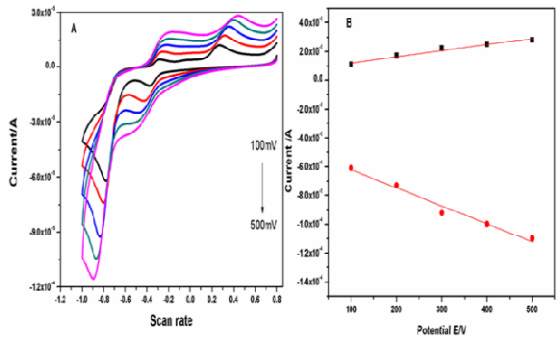

Fig 5. (A) Cyclicvoltammogram of 1mM 2-Np in 0.1M PBS (pH 6.0) at different scan rate from 100-500mV s^{-1} at KV-HA modified GCE . **(B)** Plot of anodic and cathodic peak current vs. scan rate.

3.4. Supporting Electrolyte and Effect of pH

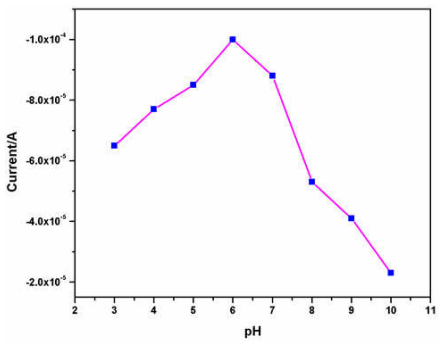

Fig 6. Effect of pH on CV peak currents of 1mM of 2-Np in 0.1M PBS at KV-HA modified GCE at the scan rate 50 mV s^{-1}.

2-Np has different electrochemical behaviors in different electrolytes. The effects of electrolytes, such as HCl, phosphate buffer, acetate buffer, and KNO_3, on reduction peak currents of 2-Np were studied and the results have shown that the 2-Np have the best electrochemical responses in 0.1M PBS. When the measurements were performed in this electrolyte, well cathodic peaks with the lowest background current were obtained. Moreover, the electrochemical methods depend on pH variation studies on the electrolyte. Here, we change pH from 3 to 10 by adding HCl and NaOH for PBS electrolyte. The best peak current appeared in pH 6.0. When high pH, peak current will be decreased and destroy the shape of a peak due to the lack of protons H^+. At low pH, current decreases due to the immobilization or dissolving hydroxyapatite in a highly acidic medium. Fig 6 shows Corresponding calibration plot for different pH studies.

3.5 Determination of 2-Nitrophenol by using DPV

Electrochemical successive determination of 2-Nitrophenol by using Kv-HA modified glassy carbon electrode in DPV technique in above optimized parameter. Fig 7 (A) shows the continuous adding aliquot from the stock of solution 2-Nitrophenol. The peak current increasing linear due to the increasing the concentration of 2-Nitrophenol from 0.2µM to 60 µM. Fig 7(B) shows a calibration plot for DPV, the resulting plot linear with the correlation coefficients R^2=0.99714. The limit of detection (LOD) was found to be 90nM.

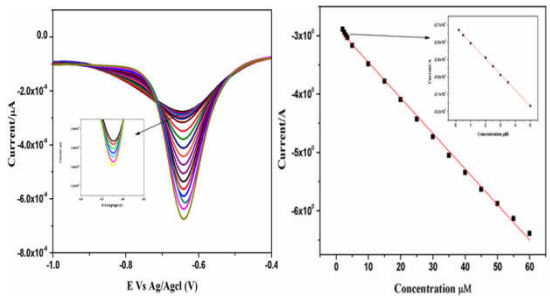

Fig. 7 (A) DPV of 2-Np from 0.2 to 60µM in 0.1M PBS (pH 6) using KV-HA modified glassy carbon electrodes. (B) Corresponding linear calibration plot.

3.6 Selectivity, Reproducibility and Stability

A significant benefit of KV-HA Nps is good selectivity for the 2 phenol compounds such as 3-Np, 4-Ap, Hydroquinone, Catechol ,and some other compounds Nitrite, Mg^{2+}, K^+, Na^+. The presence of this competing molecule did not have an obvious influence on the current response of 2-NP on the KV-HA/GCE, it

demonstrates that the KV-HA sensor can be used directly in detecting 2-NP in complex solutions. However, 4-Np and 2-Np are same structure, it's highly overlaps each other in cathodic peaks, So oxidation signal are over and over again used for simultaneous determination of 4-Np and 2-Np [17-22].But, interestingly our KV-HA /GCE can split the cathodic peak potential of 4-Np and 2- Np. Fig 6(B) shows the analytical performance of the KV-HA/GCE toward varying concentration from 3µM to 40µM of 2-Np in the presence of high concentration 50µM of 4-Np by DPV measurements. As can be seen two distinct cathodic peaks appear for 2-Np and 4-Np with a clean separation potential (110mV) and the very clear signal should be observed .This result indicates KV-HA/GCE is the best feasible electrode for a 2-Np sensor in also the presence of 4-Np. (Fig 8 (C) shows the corresponding linear fit graph. To characterize the reproducibility of the KV-HA/GCE, 15 consecutive measurements were done for 1 mM 2-Np at the same electrode. The results showed a relative standard deviation (RSD) of 2.97%, revealing that the KV-HA /GCE sensing device had good reproducibility. The stability of the KV-HA/GCE sensing device was also examined by measuring CV peak current of 1mM 2 -NP after storing the electrode for 10 days. The KV-HA/GCE sensing device retained 91% of the original values, demonstrating that the sensing device had better long- term stability.

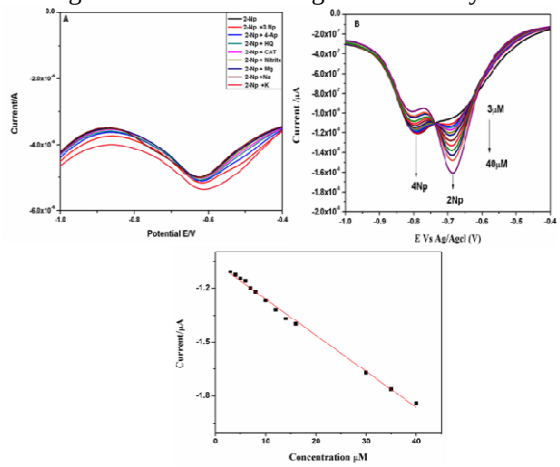

Fig. 8 (**A**) DPV responses of Influence of coexisting substances for 2-NP in KV-HA /GCE 0.1MPBS pH 6.0. (**B**) DPV responses of varying concentration from 3µM to 40µM of 2-Np in presence of 50µM of 4-Np on KV-HA GCE. (**C**) Corresponding linear fit for 2-Np on KV-HA GCE.

Table 1 : Comparison of different chemically modified electrodes for the determination of 2-Np with KV-HA/GCE

Electrode	Method	Linear	Detection	References
		Range	Limit	
Poly(p-ABSA)/GE	SDV	3-800 μM	0.28 μM	[1]
CD-RGO/GCE	DPV	7.0-64 μM	0.14 μM	[21]
CD-GNs/GCE	DPV	5–400 μM	0.3 μM	[23]
Mg/Fe-LDH/GCE	Amperometric	5-560 μM	4 μM	[24]
PPI-AuNP/GCE	SWV	5-645 μM	0.45 μM	[25]
Kv-HA/GCE	DPV	0.2-60 μM	0.09 μM	Present work

Table 2 : Determination of 2-Np in tap water samples using KV-HA/GCE.

Sample	Added(µMol/L^{-1})	Founded(µMol/L^{-1})	Recovery (%)
2 Np	10	9.7	97
2 Np	15	15.2	100.2
2 Np	20	19.8	98

3.7 Analytical Applications

Tap water samples were selected for real sample analysis using the standard addition method. The developed sensor method (KV-HA/GCE) was applied to test 2-Np in tap water samples. 2-Np solution was added into tap water and analyzes were done without any prior treatment. The results were presented in Table 2. It can be clearly observed that a good recovery was obtained, suggesting the proposed method could be effectively used for the determination of 2-Np -in real samples.

IV. CONCLUSION

In this work, we developed medicinal plant extracts Eucalyptus(ES-HA),Ocimum basicillium (TS-HA), Plectranthus amboinicus (KV-HA), Justica Adathoda (AD-HA), based hydroxyapatite nanopartilces were synthesized , and successfully applied to detect 2-Nitrophenol . Among them , Plectranthus amboinicus assisted HA Nps (KV-HA/GCE) provides better sensitivity, reproducibility and good selectivity towards 2-Nitrophenol. The cathodic peak current increase linearly with increasing the concentration of 2-Nitrophenol from 0.2µM to 60µM with the lowest limit of detection 90nM. Apart from that KV-HA modified GCE is best selective material for 2-Nitrophenol determination in the presence of high concentration of 4-Np .This

new nontoxic material is directly employed to tap water for real sample analysis with good results.

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